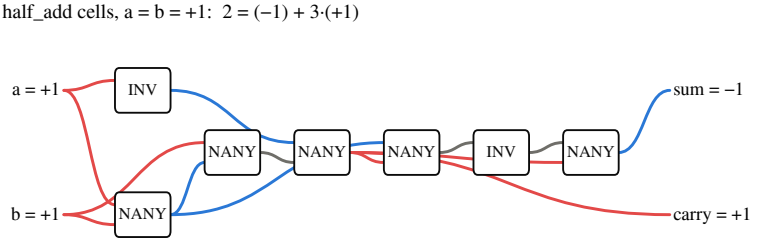
**(a)**

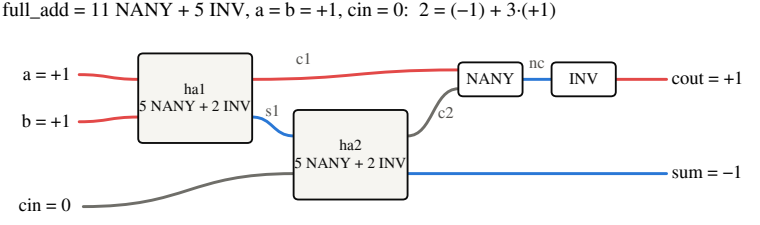
```
module half_add(input trit a, b,
                output trit sum, carry);
  wire trit t, u, na, d, nd;
  nany g_t (t, a, b);
  nany g_u (u, b, t);
  inv  g_na (na, a);
  nany g_c (carry, na, u);
  nany g_d (d, t, carry);
  inv  g_nd (nd, d);
  nany g_s (sum, nd, carry);
endmodule

module full_add(input trit a, b, cin,
                output trit sum, cout);
  wire trit s1, c1, c2, nc;
  half_add ha1 (.a(a),  .b(b),   .sum(s1),  .carry(c1));
  half_add ha2 (.a(s1), .b(cin), .sum(sum), .carry(c2));
  nany g_nc (nc, c1, c2);
  inv  g_cout (cout, nc);
endmodule
```

**(b)**




**(c)**




net at −1 net at 0 net at +1